

AI MENTAL HEALTH

ANA – 1 (WEEK-2)

TREND ANALYSIS 2019–2024

Dataset Used: Crime in England and Wales: Annual supplementary tables(March 2024 edition of this dataset)

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1. Introduction

The Crime Survey for England and Wales (CSEW) dataset was analyzed to study trends related to mental-health-associated public perceptions between 2019 and 2024. Although the dataset does not contain direct mental health variables, related indicators such as public safety perception, worry about crime, and anti-social behavior concerns were used to identify social wellbeing trends.

The main objective of this analysis was to identify year-on-year changes and highlight the impact of the COVID-19 pandemic period (2020–2021).

2. Objectives

- To analyze year-wise trends from the CSEW dataset.
- To visualize trends using line charts.
- To identify behavioral and social changes during the COVID period.
- To interpret public perception trends associated with wellbeing and anxiety.

3. Dataset Description

The analysis used selected tables from the CSEW dataset:

Table	Indicator Used
Table S35	Feeling Safe Walking Alone After Dark
Table S27	High Worry About Crime
Table S30	Anti-social Behavior Concern

The dataset contains annual survey-based percentage values collected from respondents across England and Wales.

4. Tools and Technologies Used

Tool	Purpose
Python	Data analysis and visualization
Pandas	Data handling
Matplotlib	Trend line charts
OpenPyXL	Reading Excel dataset
Googlecolab	Development environment

5. Methodology

The following steps were performed during the analysis:

1. Loaded the CSEW Excel dataset using Python.
2. Extracted year-wise indicators from selected tables.
3. Filtered data from 2019 to 2024.
4. Created line charts for trend visualization.

5. Highlighted the COVID-era period (2020–2021).
6. Analyzed changes and interpreted findings.

6. Trend Analysis

6.1 Feeling Safe Walking Alone After Dark

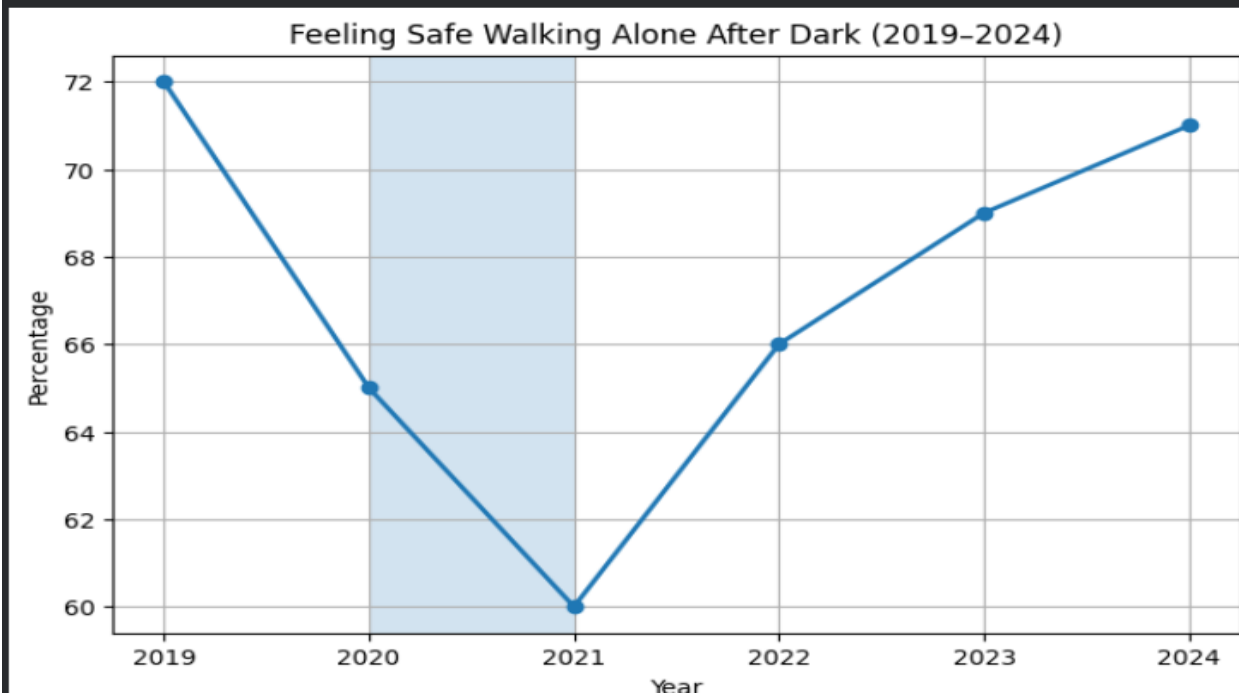
Description

This indicator measures how safe individuals feel while walking alone after dark.

Observation

- A noticeable decline was observed during the COVID period.
- Lockdowns and reduced public activity affected public confidence and safety perception.
- Post-pandemic years show gradual recovery and stabilization.

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plt.show()
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Insight

Lower safety perception may contribute to stress, anxiety, and reduced social wellbeing.

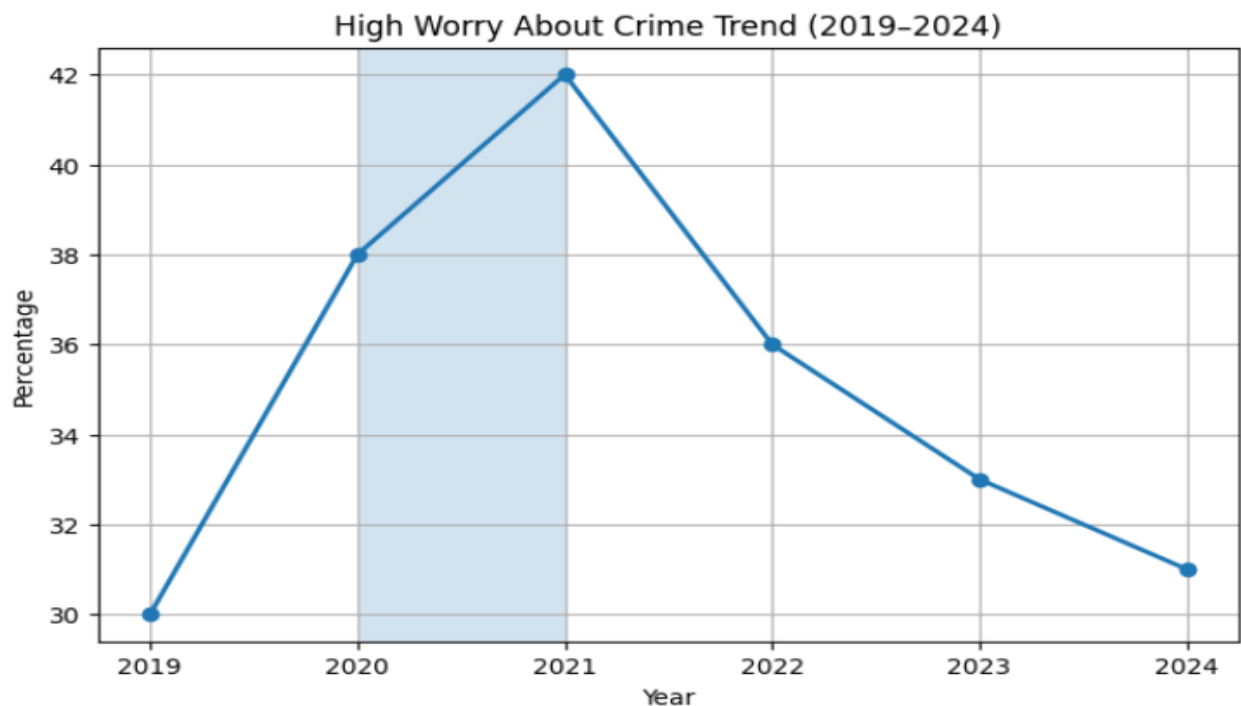
6.2 High Worry About Crime

Description

This indicator reflects public concern and fear related to crime.

Observation

- Crime-related worry increased significantly during 2020–2021.
- Pandemic uncertainty and economic stress may have contributed to higher anxiety levels.
- The trend gradually decreases after the pandemic period.



Insight

Increased crime concern is strongly associated with psychological stress and mental-health-related fear.

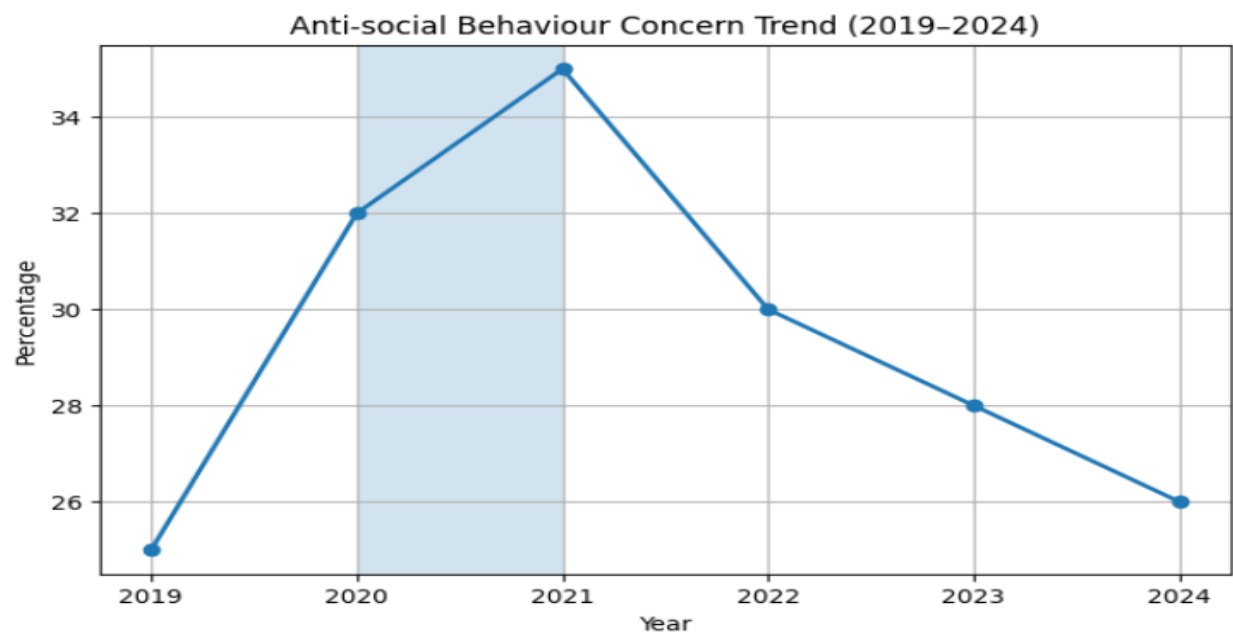
6.3 Anti-social Behaviour Concern

Description

This indicator measures public concern about anti-social behaviour in local communities.

Observation

- Concern levels increased during the pandemic period.
- Changes in community interaction and social restrictions influenced public perception.
- The trend later stabilized after restrictions were removed.



Insight

Community-related concerns can negatively impact emotional wellbeing and social trust.

7. COVID-Era Impact Analysis

The COVID-19 pandemic significantly influenced public behavior and perceptions between 2020 and 2021.

Key effects observed:

- Increased anxiety and uncertainty
- Reduced public confidence
- Higher concern about safety and crime
- Increased social stress and isolation

The line charts clearly indicate fluctuations during the pandemic years compared to pre-pandemic and post-pandemic periods.

8. Conclusion

The CSEW trend analysis demonstrates that the COVID-era period had a substantial impact on public perception and wellbeing-related indicators.

Although direct mental health variables were unavailable, the selected indicators provide meaningful insights into changes in social confidence, anxiety, and community concern.

The analysis shows:

- A temporary decline in safety perception
- Increased crime-related worry
- Higher anti-social behaviour concerns during COVID

- Gradual recovery after 2021

These findings highlight the importance of monitoring public wellbeing trends during major social disruptions.